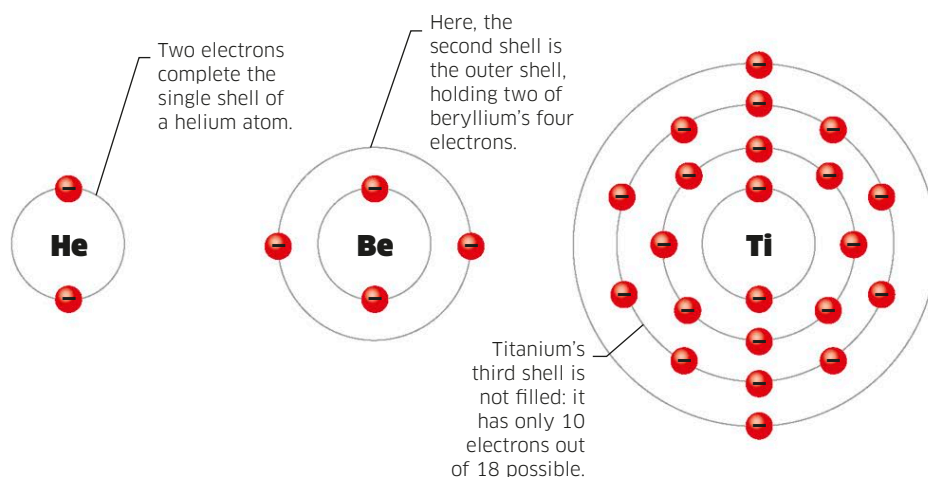


Electrons and electron shells

An atom's electrons are arranged around the nucleus in shells. Each shell can hold a certain number of electrons before it is full: the inner shell can hold 2, the next shell 8, the third one 18, and so on. The heaviest atoms, with large numbers of electrons, have

seven shells. Atoms that don't have full outer shells are unstable. They seek to share, or exchange, electrons with other atoms to form chemical compounds. This process is known as a chemical reaction. Atoms with a filled outer shell are stable, and therefore very unreactive.



Helium

The gas helium has the atomic number 2. All its atoms have two electrons, which is the maximum number the first shell can hold. With a full outer shell, helium atoms are very unreactive.

Beryllium

The second shell of an atom can hold up to eight electrons. The metal beryllium (atomic number 4) has a filled inner shell, but only two electrons in its outer shell, making it quite reactive.

Titanium

The metal titanium (atomic number 22) has four shells. It has two electrons in its outer shell, even though the third shell is not full. It is quite common for metals to have unfilled inner shells.

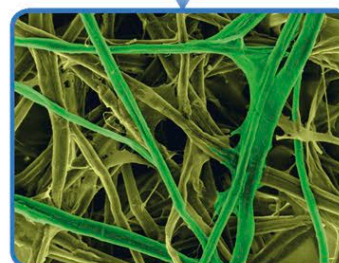
Atoms and matter

It is difficult to imagine how atoms make the world around you. Everyday objects don't look as if they consist of tiny round bits joined together: they look continuous. It can help to zoom in closer and closer to an everyday material, such as paper, to get the idea.



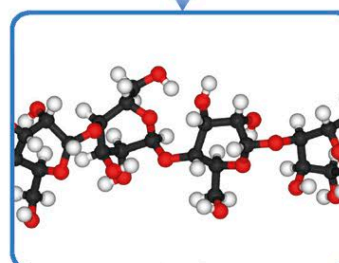
Paper

Paper is made almost entirely of a material called cellulose, which is produced inside plant cells, usually from trees. Cellulose is hard-wearing and can absorb inks and paints.



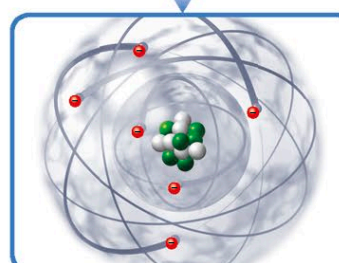
Cellulose fiber

Cellulose forms tiny fibers, each about one thousandth of a millimeter in diameter. The fibers join together, making paper strong and flexible.



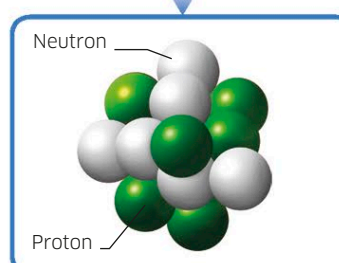
Cellulose molecule

Each cellulose fiber is made of thousands of molecules. A cellulose molecule is a few millionths of a millimeter wide. It is made of atoms of different elements: carbon (black), oxygen (red), and hydrogen (white).



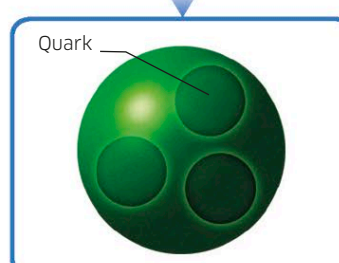
Carbon atom

A typical cellulose molecule contains a few thousand carbon atoms. Each carbon atom has six electrons that form bonds with atoms of the other two elements.



Nucleus

Most of the carbon atom is empty space. Right at the center, about one trillionth of a millimeter across, is the nucleus, made of six protons and six neutrons.



Quarks

Each particle in the nucleus is made of even smaller particles, called quarks. Each proton—and each neutron—is made of three quarks, held together by particles called gluons.

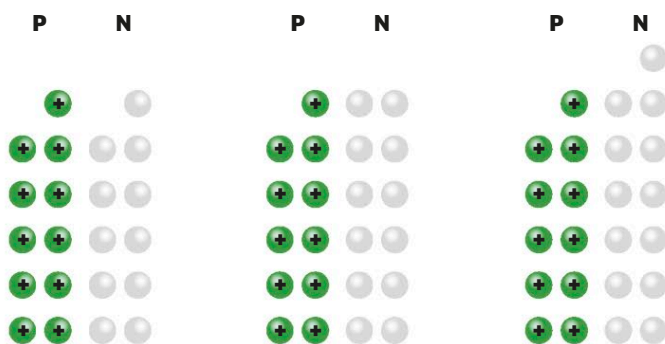
Atomic mass and isotopes

The mass of an atom is worked out by counting the particles of which it is made. Protons and neutrons are more than 1,800 times heavier than electrons, so scientists only take into account those heavier particles, and not the electrons. All atoms of a

particular element have the same number of protons, but there are different versions of the atoms, called isotopes, that have different numbers of neutrons. The relative atomic mass of an element is the average of the different masses of all its atoms.

Isotopes of sodium

All atoms of the element sodium (atomic number 11) have 11 protons, and nearly all have 12 neutrons. So the relative atomic mass is very close to 23, but not exactly.



Sodium-22

The sodium isotope with 11 neutrons in its atoms has a mass of 22.

Sodium-23

Sodium-23, the most common sodium isotope, has 11 protons and 12 neutrons.

Sodium-24

This sodium isotope has a mass of 24: 11 protons and 13 neutrons.